INCH-POUND

MIL-V-23151B(SHIPS)
AMENDMENT 4
3 December 1990
SUPERSEDING
AMENDMENT 3
26 April 1988

MILITARY SPECIFICATION

VOLTMETER, EXPANDED SCALE SWITCHBOARD TYPE
(NAVAL SHIPBOARD USE)

This amendment forms a part of MIL-V-23151B(SHIPS), dated 10 April 1967, and is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.1, 1.2, and 1.2.1: Delete and substitute:

"1.1 Scope. This specification covers high accuracy, shockproof switchboard mounted, expanded scale voltmeters.

1.2 Classification.

1.2.1 Type designation. The type designation of voltmeters shall be in the following form, and as specified (see table II and 6.2):

MR	49	W	115	AF	H
Component (1.2.1.1)	Style (1.2.1.2)	Color scheme (1.2.1.3)	Center-scale value (1.2.1.4)	Kind of current (1.2.1.5)	Shockproof feature (1.2.1.6)"

PAGE 2

1.2.1.4: Add the following:

"Center-scale value | Scale range

115	110-120
120	115-125
208	198-218
220	210-230
440	420-460
450	430-470"

AMSC N/A FSC 6625
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PAGE 3

3.4.2.1, line 2: Delete "No. 11" and substitute "No. 111".

3.4.3: Delete and substitute:

"3.4.3 Expanded scale circuit. The expanded scale circuit components shall be encapsulated in material which shall cover all components and shall be in contact with the interior of the meter barrel extension or other case or container of the circuit."

PAGE 4

3.6: Delete "3.22" and substitute "3.21".

PAGE 5

3.15: Delete and substitute:

"3.15 Shock (high-impact). When voltmeters are tested in accordance with 4.3.14 no screws, bearings, pivots, windows, or other parts shall become loosened, cracked, (except as specified herein), or unduly damaged. Minor cracking or distortion of pivots will be permissible only when conformance to the post shock performance has been demonstrated. Cracking of glass windows will be permissible only when readability of the complete scale is retained and no fragment has become dislodged. The permanent change in indication based on calibration made after resetting the pointer to zero shall not exceed the value specified (see table II)."

PAGE 8

4.3.3, line 3: Delete "45'" and substitute "25'".

PAGE 10

5.1: Add "(see 5.3)."

PAGE 11

5.2.1.1 and 5.2.2: Add "(see 5.3)."

Add the following paragraphs:

"5.3 Cushioning, filler, dunnage, and wrapping materials.

5.3.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for materials destined for shipboard installation/stowage.

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5.3.2 Level C preservation-packaging and packing. When loose-fill type materials are used for packaging and packing applications such as cushioning, filler, and dunnage, all containers (unit, intermediate, and shipping) shall be marked or labelled with the following information:

"CAUTION

Contents cushioned with loose-fill material.
Not to be taken aboard ship.
Remove and discard loose-fill material.
If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material."

5.3.3 Cushioning, filler, dunnage, and wrapping materials selected, whenever available, shall exhibit improved performance for resistance to fire."

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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